EXECUTIVE SUMMARY · FINTECH ANALYTICS

UPI Transaction Fraud & Risk Analytics

End-to-end fraud analytics across 50,000 UPI transactions | Python · PostgreSQL · Excel · Power BI

50,000	3.25%	₹67.5L	17
TRANSACTIONS ANALYZED	OVERALL FRAUD RATE	TOTAL FRAUD LOSS	SQL QUERIES WRITTEN

1. PROBLEM STATEMENT

Digital payment platforms process millions of UPI transactions daily, and fraud grows alongside that volume. Without a centralized, data-driven view of where fraud concentrates — by city, merchant, channel, or time of day — risk teams cannot prioritize investigations or measure whether controls are working. This project builds that view end-to-end: from raw transaction data to an executive-ready dashboard answering *where fraud is happening, who is most at risk, and what the business should do about it.*

2. APPROACH

Stage	Tool	Output
Data Generation	Python (Faker, NumPy)	50,000 synthetic UPI transactions across 5 related tables
Cleaning & EDA	Pandas, Matplotlib, Seaborn	12 business questions answered with charts and insights
Database & Analysis	PostgreSQL	Normalized schema + 17 business SQL queries (CTEs, window functions)
Operational Reporting	Excel (Power Query)	Row-level dashboard for day-to-day fraud investigation
Executive Reporting	Power BI	3-page interactive dashboard for management decision-making

3. KEY FINDINGS

#	Finding
1	Fraud peaks between **12 AM–4 AM**, when monitoring is lowest — a clear window for step-up authentication.
2	**Debit Card** has the highest channel fraud rate (3.73%); Wallet is the safest channel.
3	The **top 10 riskiest merchants** show fraud rates of 7.4%–9.4% — roughly 2–3x the platform average.
4	**Unverified merchants** show a modestly higher fraud rate (3.42%) than verified merchants (3.22%).
5	**KYC status alone** is not a strong fraud predictor — behavioral signals matter more.
6	New and established accounts show **similar fraud rates** — fraud is not limited to new sign-ups.

4. RECOMMENDATIONS

Finding	Recommended Action
Late-night fraud concentration (12 AM–4 AM)	Apply step-up authentication (OTP) on large transactions in this window
Debit Card has the highest channel fraud rate	Add extra verification for high-value Debit Card transactions
Top 10 merchants drive disproportionate fraud loss	Flag for manual review; consider reduced settlement limits
Unverified merchants carry higher exposure	Tighten onboarding verification before activating new merchants
Payment failures cause measurable revenue leakage	Investigate top failure reasons (e.g. server timeouts) as an engineering priority

5. DELIVERABLES

- 2 Jupyter notebooks (data generation, EDA)
- Normalized PostgreSQL schema + 17 SQL queries
- Excel operational dashboard
- 3-page Power BI executive dashboard
- Full project report with screenshots

6. SKILLS DEMONSTRATED

- End-to-end data pipeline design
- Feature engineering for fraud detection
- Advanced SQL (CTEs, window functions)
- Dual dashboarding (operational + executive)
- Business-first analytical thinking

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